



Mining Proportional Patterns in Online Retail Data

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1 Introduction

Frequent Itemset Mining (FIM) has long been a cornerstone of market basket analysis, providing retailers with the ability to discover hidden associations between products purchased together. The evolution of these techniques led to the development of the FP-Growth algorithm, which revolutionised the field by adopting a divide-and-conquer approach that avoids the prohibitively expensive candidate generation typical of earlier methods like Apriori. However, in the era of massive online retail datasets, two primary challenges emerge: computational scalability and the inherent limitation of binary data.

1.1 The Challenge of Scale and the Long Tail

Modern retail databases often reach sizes where memory and computational costs on a single machine become unsustainable. Furthermore, web-based retail data frequently exhibits a “long-tail” distribution, where a massive number of items have low individual support but represent a significant portion of overall revenue when considered collectively. To address this, the Parallel FP-Growth (PFP) framework was introduced, enabling the distribution of mining tasks across a cluster of machines using a five-step MapReduce-based pipeline. PFP achieves virtually linear speedup by sharding the transaction database into independent tasks, eliminating the need for inter-machine communication during the mining process.

1.2 Moving Beyond Binary Mining

Despite the efficiency of PFP, traditional FIM frameworks are largely limited to the binary presence or absence of items, effectively ignoring the purchased quantities. In a retail context, understanding the exact proportions of items bought together (e.g., a “2-for-4” ratio of related products) provides deeper insights for bundle recommendations, targeted upselling, and inventory management.

1.3 The Proposed Extension: Proportional Analysis

This project proposes a structural extension to the PFP framework designed to extract proportional quantitative relationships. By implementing a custom Bijective Token Indexing phase, we adapt the PFP distributed engine to handle quantitative tuples without modifying its core prefix-tree logic. The primary contribution of this work lies in the post-mining analytical layer: we apply a Greatest Common Divisor (GCD) decomposition to identify the Canonical Shape of frequent patterns — the irreducible proportional ratio of items. This logic allows us to bridge the gap between tree-based mining and structural graph analysis, culminating in a Proportional Relationship Graph. By applying transitive reduction to these results, we eliminate redundant informational pathways, providing an optimised, hierarchical atlas of consumer purchase trajectories that highlights the structural evolution of consumer behaviour.

2 Formal Methodology: Quantitative Itemsets, Canonical Shapes, and Scaling Factors

To rigorously analyse co-occurrences in retail data while accounting for purchased quantities, we establish a formal mathematical framework [1, 2]. Let $I = \{i_1, i_2, \dots, i_m\}$ be the universe of distinct items [3]. A **quantitative itemset** X is defined as a set of ordered pairs representing items and their specific purchased quantities: $X = \{(i_j, q_j) \mid i_j \in I, q_j \in \mathbb{Z}^+\}$ [3].

For any frequent quantitative pattern $P \in \mathcal{P}$ characterised by the quantity set $\{q_1, q_2, \dots, q_k\}$, we define the **Scaling Factor** α as the greatest common divisor of those quantities: $\alpha = \gcd(q_1, q_2, \dots, q_k)$ [4]. This extraction of a common scalar is theoretically grounded in the **Prefix Path Property**, which states that for any node a_i in a path P within an FP-tree, the frequency count of every node in its prefix subpath must carry the same count as node a_i [5]. In a quantitative context, this property suggests that items appearing in

a single path often co-occur in fixed ratios, justifying the use of a scaling factor to represent the magnitude of the transaction [4, 5].

Consequently, each pattern P can be uniquely decomposed into its **Canonical Shape** S , which represents the irreducible proportional ratio of the items [4]. The shape S is derived by normalising each quantity in the pattern by the scaling factor:

$$S = \left\{ \left(i_j, \frac{q_j}{\alpha} \right) \mid (i_j, q_j) \in P \right\} \quad (1)$$

By sorting these items in a fixed lexicographical order, we ensure a unique representation for each structural relationship [4, 6, 7]. Under this formulation, any extracted frequent pattern is represented as a scalar multiple of its base structure: $P = \alpha \cdot S$ [4].

3 Distributed Architecture

The massive scale of modern retail transactions necessitates a distributed computing approach to avoid memory overflows and prohibitive computational costs [3, 4]. Our architecture leverages the **Parallel FP-Growth (PFP)** framework [5], adapted to process quantitative itemsets through a multi-stage MapReduce-based pipeline. This architecture ensures virtually linear speedup by eliminating inter-node communication during the mining process [3, 6].

3.1 Bijective Token Indexing and Vocabulary Discovery

Standard FIM algorithms are designed to process discrete tokens rather than complex tuples. To bridge this gap without modifying the underlying prefix-tree logic, we implement a **Bijective Token Indexing** phase (adapted from PFP’s Parallel Counting step [7, 8]).

- **Sharding:** The raw transactional database DB is partitioned into P independent shards across the cluster [1].
- **Parallel Counting:** Each mapper discovers the local vocabulary of $(item, quantity)$ pairs. The MapReduce infrastructure aggregates these counts to generate a global **F-List** [7, 9].
- **Tokenisation:** Every unique pair is mapped to a synthetic integer identifier $ID \in \mathbb{Z}^+$. This standardises the input for the PFP engine, ensuring that a purchase of “2 colas” is treated as a distinct frequent entity from “1 cola” [10].

3.2 Workload Partitioning and the Group List

A critical challenge in distributed FP-Growth is balancing the computational load. PFP addresses this by dividing the $|I|$ items from the F-List into Q independent groups, creating a **Group List (G-list)** [7]. Each group is assigned a unique *group-id* (gid). By setting Q significantly higher than the number of physical

machines (P), we ensure that if one node finishes its mining task early, it can immediately process the next group, preventing straggler nodes from bottlenecking the pipeline [11, 12].

3.3 Independent Parallel Mining

The core of the architecture lies in the **Parallel FP-Growth Step** [13]. This stage transforms the database into group-dependent transactions:

1. **Mapper Phase:** Each mapper reads the G-list and outputs group-dependent transactions. For each item in a transaction, it emits a key-value pair where the key is the *gid* and the value is the prefix path of that item [13, 14].
2. **Reducer Phase:** Reducers receive independent shards of group-dependent transactions. Each reducer builds a **local FP-tree** and mines it recursively to extract patterns [15, 16].

Because the shards are mathematically independent, there is no need for synchronisation between workers during the recursive tree-growth process, which is the primary reason for PFP’s high scalability [3, 5].

3.4 Max-Heap Aggregation

To manage the potentially combinatorial number of patterns, each reducer maintains a **Max-Heap (HP)** of size K [17, 18]. Instead of emitting all discovered patterns, the reducer retains only the top- K most supported patterns per item. This filtered output is aggregated in the final step to produce the global top- K result, which serves as input for the Proportional Graph Construction phase.

4 Implementation & Algorithm Design

The proposed computational pipeline is engineered to bridge the gap between distributed statistical mining and structural relationship modelling. The implementation is divided into three functional phases, each addressing a specific challenge of quantitative big data processing within an Apache Spark environment.

4.1 Phase 1: Distributed Dataset Transformation

Standard Frequent Itemset Mining (FIM) engines, including the PFP framework, are designed to process discrete, non-quantitative tokens. To preserve purchased quantities without altering the underlying prefix-tree logic, we implement a *Bijective Token Indexing* phase.

This phase executes a distributed transformation where every unique $(item, quantity)$ tuple discovered across the database is assigned a continuous synthetic identifier $ID \in \mathbb{Z}^+$. By leveraging Spark’s

`zipWithIndex` operation, we effectively perform the vocabulary discovery described in Step 2 of the PFP framework. This pre-processing ensures that quantitative variations (e.g., `cola:1` vs. `cola:2`) are treated as distinct frequent entities, maintaining the granularity required for proportional analysis.

Algorithm 1: Bijective Token Indexing

Input: Transaction database DB with $(item, quantity)$ tuples
Output: Transformed database DB' with integer IDs
parallel for each transaction $T \in DB$ *extract* all pairs (i, q) ;
aggregate global vocabulary of unique (i, q) pairs;
assign unique $ID \in \mathbb{Z}^+$ *using* `zipWithIndex`;
parallel for each transaction $T \in DB$ *replace* each (i, q) with corresponding ID ;
return DB' ;

4.2 Local FP-Growth Mining

The local FP-Growth procedure was implemented from scratch rather than delegating to MLlib’s built-in `FPGrowth`, which operates on discrete tokens and cannot handle the quantitative tuples produced by the Bijective Token Indexing phase. The implementation centres on a custom `FPTree` class that maintains an explicit header table — a dictionary mapping each item to the list of its nodes in the tree — enabling direct traversal during conditional tree construction without a full tree scan.

A correctness-critical design choice lies in the conditional pattern base construction: every prefix node is assigned the count of the conditioning item rather than its own count. This ensures that the support of any pattern discovered in the conditional tree is bounded by the frequency of the conditioning item, as required by the Prefix Path Property. Deviating from this — for instance, preserving original node counts — would inflate support estimates and produce spurious frequent patterns.

When a conditional tree reduces to a single path, all nodes inherit the same count from the conditioning item, making support-based ranking among combinations redundant. Our implementation exploits this by enumerating combinations in strictly ascending order of cardinality, retaining only the first K . Beyond correctness, this ordering is a deliberate system-level trade-off: shorter patterns incur lower serialisation overhead during the `combineByKey` shuffle, reducing inter-node communication cost. This is particularly relevant in long-tail datasets where a disproportionate number of conditional trees reduce to single paths.

Algorithm 2: TOPKCOMBINATIONS — cardinality-ascending enumeration

Input: Single-path nodes $\mathcal{P}$, budget K
Output: First K combinations in ascending cardinality order
 $count \leftarrow 0$;
for $r \leftarrow 1$ **to** $|\mathcal{P}|$ **do**
 for each combination C **of size** r **from** $\mathcal{P}$ **do**
 if $count \geq K$ **then return**;
 yield C ;
 $count \leftarrow count + 1$;

Pattern volume is controlled through a Max-Heap of size K , maintained locally by each worker. When the heap exceeds capacity, the lowest-support pattern is evicted. Local heaps are merged in the aggregation phase via `combineByKey`, which ensures the merging is associative and commutative — a requirement for correct distributed reduction. The choice of K directly governs the precision-scalability trade-off: a larger K retains more structural information but increases shuffle volume and memory pressure on the aggregator nodes.

Algorithm 3: Local Top- K FP-Growth

Input: FP-tree T , minimum support ξ , suffix σ , heap size K
Output: Top- K frequent patterns with support
if T *reduces to a single path* $\mathcal{P}$ **then**
 $H \leftarrow []$;
 for each $C \in \text{TOPKCOMBINATIONS}(\mathcal{P}, K)$ **do**
 $sup \leftarrow C[0].count$;
 if $sup \geq \xi$ **then**
 append $(sup, \text{items}(C) + \sigma)$ to H ;
 return H ;
 $\mathcal{H} \leftarrow []$;
for each $(item, nodes)$ *in header table of* T **do**
 $sup \leftarrow \sum_{n \in nodes} n.count$;
 if $sup < \xi$ **then skip**;
 $\pi \leftarrow (item,) + \sigma$;
 $\mathcal{H} \leftarrow \text{ADDTOHEAP}(\mathcal{H}, (sup, \pi), K)$;
 $T' \leftarrow \text{CONDITIONALTREE}(T, item, \xi)$;
 if T' *is non-empty* **then**
 $\mathcal{H}' \leftarrow \text{TOPKFPKGROWTH}(T', \xi, \pi, K)$;
 $\mathcal{H} \leftarrow \text{MERGEHEAPS}(\mathcal{H}, \mathcal{H}', K)$;
return $\mathcal{H}$;

4.3 Phase 2: Parallel Pattern Extraction

With the database standardised into ID sequences, the system triggers the Parallel FP-Growth engine. The transaction database is divided into independent shards, allowing each worker to execute local mining tasks without inter-machine communication. Each shard is processed by a dedicated reducer that builds a local FP-tree and invokes Algorithm 3 recursively.

Once mining is complete, an inverse mapping M^{-1} is applied to restore the structural $(item, quantity)$ tuples for final analysis.

Algorithm 4: Parallel FP-Growth with Top- K Filtering

Input: Transformed database DB' , parameter K

Output: Top- K frequent patterns

partition DB' into independent shards;

parallel for each shard build local FP-tree;

for each discovered pattern P via Algorithm 3 **do**

insert into Max-Heap HP of size K based on support;
prune if $|HP| > K$;

aggregate all local heaps;

apply inverse mapping M^{-1} to recover $(item, quantity)$ tuples;

return Top- K patterns;

4.4 Phase 3: Proportional Graph Construction

The final phase transforms the flat list of frequent quantitative patterns into a hierarchical Directed Acyclic Graph (DAG) by exploiting the mathematical structure of their scaling factors.

4.4.1 GCD Decomposition

For each extracted pattern, the system computes the Scaling Factor α and the Canonical Shape S via a parallel GCD calculation. Patterns are then clustered by identical shape using Spark's `groupByKey`, which routes all patterns sharing the same structural ratio to a single worker node — ensuring that the subsequent graph construction is both locally complete and globally independent across shape groups.

4.4.2 Transitive Modulo Pruning

Within each shape group, the algorithm performs a pairwise divisibility check over the set of scaling factors $V = \{\alpha_1, \dots, \alpha_n\}$. An edge (α_i, α_j) is created if $\alpha_j \bmod \alpha_i = 0$, signifying a perfect scalar transition. Raw edge generation is then refined by transitive reduction: any direct edge (P_i, P_j) is pruned if an intermediate node P_k exists such that $P_i \rightarrow P_k \rightarrow P_j$. This eliminates informational redundancy while preserving all structurally meaningful scaling increments. The local $\mathcal{O}(|\alpha|^2)$ cost of this check is acceptable because each group is processed independently, keeping the global phase parallelisable across shape groups.

Algorithm 5: Proportional Graph Construction with Transitive Modulo Pruning

Input: Set of frequent patterns $\mathcal{P}$

Output: Directed Acyclic Graph (DAG)

parallel for each pattern $P \in \mathcal{P}$ compute

$\alpha = \gcd(q_1, \dots, q_k)$;

compute canonical shape $S = P/\alpha$;

group patterns by identical S using `groupByKey`;

parallel for each group S let $V = \{\alpha_1, \dots, \alpha_n\}$;

foreach (α_i, α_j) with $\alpha_i < \alpha_j$ **do**

if $\alpha_j \bmod \alpha_i = 0$ **then**

create edge (P_i, P_j) ;

remove (P_i, P_j) if $\exists P_k$ such that $P_i \rightarrow P_k \rightarrow P_j$;

return DAG;

4.5 Pipeline Correctness: A Worked Example

To verify the correctness of the full pipeline end-to-end, we trace the execution of Algorithms 1–5 on a minimal synthetic example designed to exercise both the linear scaling topology and the multi-parent convergence case — the two structurally distinct DAG configurations the transitive reduction must handle correctly.

The example considers two disconnected frequent structures: a three-item Beverage association with base ratio 1:2:3 (cola, juice, water) and a two-item Pancake association with base ratio 2:1 (flour, milk). Table 1 summarises the GCD decomposition of the patterns extracted by the PFP engine. Despite varying absolute quantities, each group maps to a single canonical shape, confirming that the tokenisation and inverse mapping steps preserve proportional structure correctly.

The Beverage group G_{S_1} exercises the multi-parent convergence case. The scale set $V_{S_1} = \{1, 2, 3, 6\}$ admits three raw divisibility edges from P_1 : toward P_2 , P_3 , and P_4 . The transitive reduction correctly identifies that P_4 is reachable from P_1 via both $1 \rightarrow 2 \rightarrow 6$ and $1 \rightarrow 3 \rightarrow 6$, and prunes the redundant direct edge (P_1, P_4) while preserving (P_2, P_4) and (P_3, P_4) — since $2 \nmid 3$ and $3 \nmid 2$, these represent genuinely independent scaling paths. The Pancake group G_{S_2} exercises the simpler linear case: $V_{S_2} = \{1, 2, 4\}$ yields a clean chain $P_5 \rightarrow P_6 \rightarrow P_7$ with no redundant edges. Both resulting DAGs are shown in Figure 1.

Group	Pattern	$\gcd(\cdot)$	Canonical Shape (S)	Scale (α)
G_{S_1}	P_1	$\gcd(1, 2, 3) = 1$	$\{(\text{cola}, 1), (\text{juice}, 2), (\text{water}, 3)\}$	1
	P_2	$\gcd(2, 4, 6) = 2$	$\{(\text{cola}, 1), (\text{juice}, 2), (\text{water}, 3)\}$	2
	P_3	$\gcd(3, 6, 9) = 3$	$\{(\text{cola}, 1), (\text{juice}, 2), (\text{water}, 3)\}$	3
	P_4	$\gcd(6, 12, 18) = 6$	$\{(\text{cola}, 1), (\text{juice}, 2), (\text{water}, 3)\}$	6
G_{S_2}	P_5	$\gcd(2, 1) = 1$	$\{(\text{flour}, 2), (\text{milk}, 1)\}$	1
	P_6	$\gcd(4, 2) = 2$	$\{(\text{flour}, 2), (\text{milk}, 1)\}$	2
	P_7	$\gcd(8, 4) = 4$	$\{(\text{flour}, 2), (\text{milk}, 1)\}$	4

Table 1: GCD decomposition into canonical shapes and scaling factors for the two synthetic consumer behaviours.

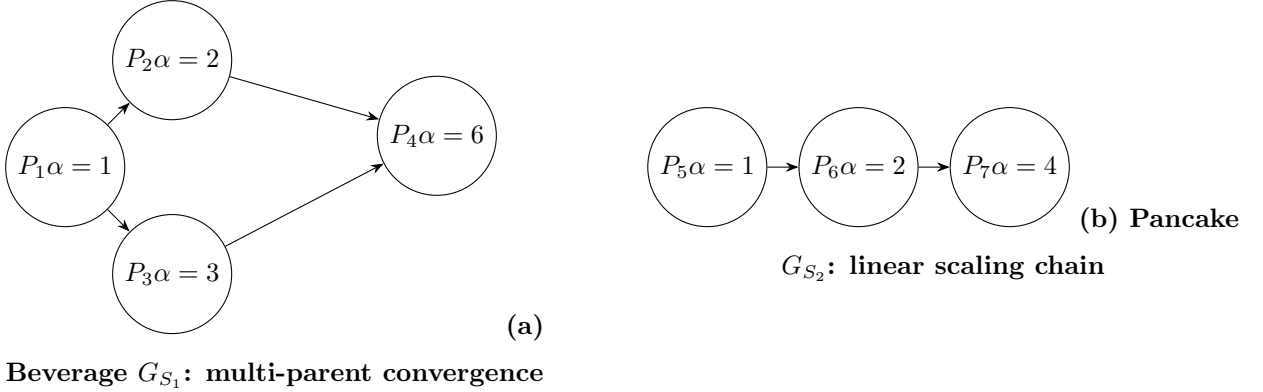


Figure 1: DAGs produced by the transitive reduction step on the synthetic example. (a) exercises multi-parent convergence; (b) exercises linear scaling. In both cases the algorithm produces the minimal edge set.

5 Experimental Evaluation

This section evaluates the computational performance and output characteristics of the proposed pipeline on the Dunnhumby *The Complete Journey* dataset. All experiments were executed locally using Apache Spark in `local[*]` mode, leveraging all available cores. The primary objectives are to characterise the scaling behaviour of the Parallel FP-Growth engine with respect to dataset size, and to quantify the overhead and effectiveness of the Proportional Graph Construction phase.

5.1 Dataset and Experimental Setup

The Dunnhumby dataset contains 2,595,732 purchase records from 2,500 households over 52 weeks, covering approximately 92,000 unique products. After grouping records by basket identifier, the dataset yields approximately 280,000 unique transactions. The quantity distribution is highly skewed: while the median purchase quantity is 1 unit, the mean is approximately 100 and the maximum reaches 89,638 units, reflecting the extreme long-tail characteristic of real-world retail data. This skew directly stresses the GCD-based canonical shape extraction, as outlier quantities can suppress the discovery of proportional relationships — a limitation discussed in Section 6. Negative quantities representing product returns are handled transparently by the absolute-value normalisation applied during tokenisation.

Experiments were conducted as a partial grid search over the following ranges: minimum support $\xi \in \{10, 15, \dots, 100\}$, heap size $K \in \{200, 400, \dots, 2000\}$,

number of groups $Q \in \{100, 200, \dots, 1000\}$, and dataset fraction $f \in \{0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.8, 1.0\}$. Each configuration was preceded by a warmup run on a 5% subsample to mitigate JVM JIT compilation bias, ensuring that measured times reflect steady-state execution rather than cold-start overhead.

Reference curves for $\mathcal{O}(n)$ and $\mathcal{O}(n^2)$ growth were calibrated by fitting a multiplicative constant c via linear regression in log-log space:

$$c = \exp\left(\frac{1}{N} \sum_{i=1}^N [\log t_i - k \cdot \log n_i]\right) \quad (2)$$

where $k=1$ for the linear reference and $k=2$ for the quadratic one. This log-space estimator assigns uniform weight to all observations regardless of sample size, avoiding the distortion of ordinary least squares in settings where the dynamic range of n spans several orders of magnitude.

5.2 Scalability of the Mining Phase

Figure 2 reports mining time as a function of sample size at the most demanding configuration tested ($\xi=10$, $K=1200$, $Q=600$). The measured times closely track the $\mathcal{O}(n)$ reference across the full range from $\sim 30,000$ to $\sim 280,000$ transactions, confirming that the PFP engine achieves near-linear scaling on this dataset. At full dataset size, the mining phase completes in approximately 72 seconds. The $\mathcal{O}(n^2)$ reference diverges markedly, ruling out super-linear growth even at the lowest support threshold where the pattern space is largest. This result is consistent with the theoretical

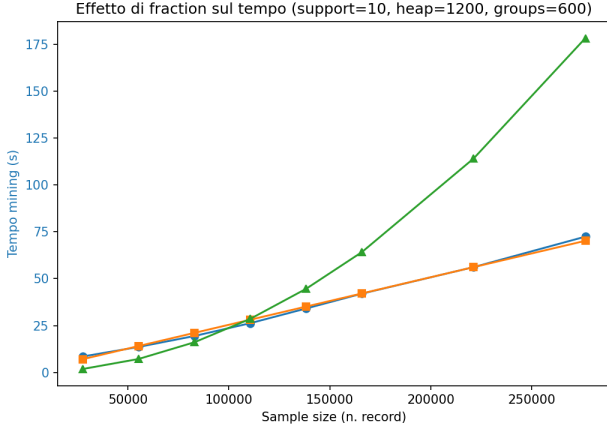


Figure 2: Mining time as a function of dataset size ($\xi=10$, $K=1200$, $Q=600$). Measured data (blue) closely follows the $\mathcal{O}(n)$ reference (orange), confirming near-linear scalability. The $\mathcal{O}(n^2)$ reference (green) diverges significantly.

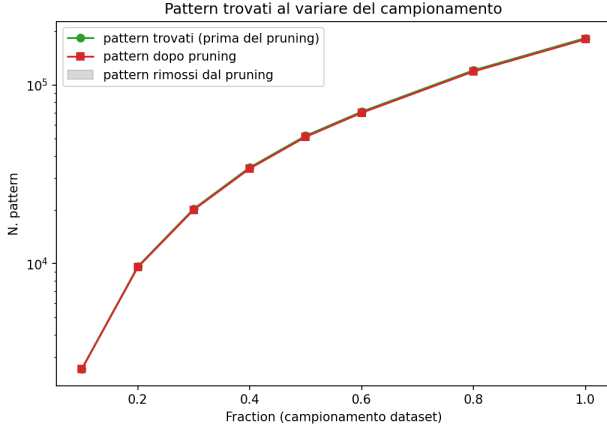


Figure 3: Pattern count before and after Transitive Modulo Pruning as a function of dataset fraction (log scale). The two curves are nearly indistinguishable across all fractions.

guarantee of PFP: because group-dependent transaction shards are mathematically independent, the recursive mining step scales with the size of each shard rather than the full dataset.

5.3 Pattern Volume and Dataset Size

Figure 3 shows the number of frequent patterns discovered as the dataset fraction increases, on a logarithmic scale. Pattern count grows from approximately 2,000 at 10% sampling to 175,000 at full dataset size. The sub-linear growth on the log scale reflects the diminishing marginal return of additional transactions once the most frequent item combinations are already saturated at smaller fractions. The near-identical gap between the pre-pruning and post-pruning curves across all fractions anticipates the finding of the following subsection: transitive reduction removes a negligible fraction of patterns, independent of dataset size.

5.4 Pruning Effectiveness

Figure 4 quantifies the effectiveness of Transitive Modulo Pruning as minimum support varies from 10 to 100. Pattern count drops sharply with increasing support — from approximately 180,000 at $\xi=10$ to fewer than 5,000 at $\xi=100$ — confirming the expected inverse relationship. The fraction of patterns removed by transitive reduction peaks at approximately 1.5% at the lowest support and decays to around 0.2% at the highest.

This low pruning rate warrants a critical interpretation. It indicates that the Dunnhumby dataset contains few multi-level scaling trajectories — few canonical shapes S for which three or more distinct scaling factors satisfy the full divisibility chain $\alpha_i \mid \alpha_j \mid \alpha_k$. This is consistent with the dataset’s quantity distribution: with the overwhelming majority of purchases involving a single unit, the GCD decomposition rarely produces the layered α structures that transitive reduction is designed to simplify. The step is therefore most valuable in datasets with richer purchasing quantity distributions, such as wholesale or manufacturing procurement data.

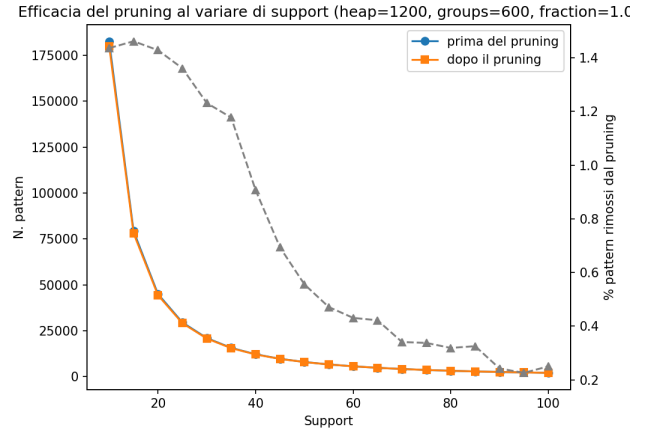


Figure 4: Pattern count before (blue) and after (orange) Transitive Modulo Pruning as a function of minimum support, with the removal percentage on the right axis (dashed). The pruning rate remains below 1.5% across all configurations.

Figure 5 reports the fraction of total execution time consumed by the alpha extraction and pruning phase. The overhead is stable in the range 4.4%–5.9% across all tested support values, with a slight upward trend at higher support. This counter-intuitive increase occurs because higher support reduces absolute mining time while the fixed operational costs of the Spark job — task scheduling, broadcast serialisation, and `groupByKey` overhead — become proportionally larger. Critically, the overhead never exceeds 6% of total execution time, validating the architectural decision to confine GCD decomposition and transitive reduction to the aggregation phase.

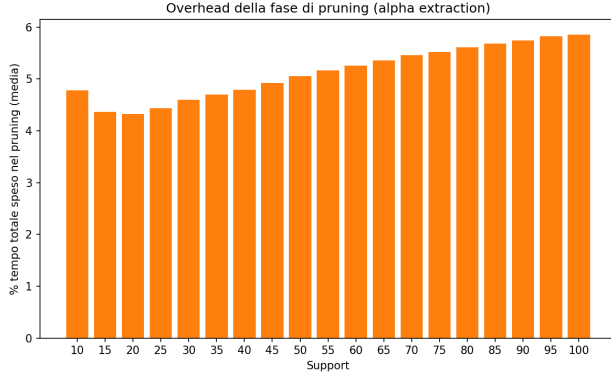


Figure 5: Fraction of total execution time spent in the alpha extraction phase as a function of minimum support. Overhead remains below 6% across all configurations.

5.5 Scalability of the Pruning Phase

Figure 6 shows the absolute execution time of the pruning phase as a function of input pattern count, on a log-log scale. Below approximately 10,000 patterns, pruning time remains sub-second with moderate variance attributable to Spark task scheduling jitter. Beyond this threshold, execution time increases more steeply and variance grows substantially, reaching 2–3 seconds at $\sim 150,000$ patterns.

The observed growth is consistent with the $\mathcal{O}(|\alpha|^2)$ pairwise divisibility check performed within each canonical shape group by `reduce_alpha_edges`: as the number of distinct scaling factors per group increases, the local quadratic cost dominates. However, because `groupByKey` ensures each group is processed independently on a single worker, this cost does not propagate globally — the pruning phase remains parallelisable across shape groups and its wall-clock time is bounded by the largest individual group rather than the total pattern count.

6 Limitations and Conclusion

6.1 Limitations and Future Work

While the proposed pipeline successfully extracts structural proportions from massive datasets, several constraints remain that offer opportunities for future research:

- **Sensitivity to Quantity Noise:** The reliance on the GCD for extracting the Scaling Factor (α) makes the model highly sensitive to noise in purchase quantities. A minor deviation in a single transaction (e.g., a user purchasing 11 units instead of 10) can prevent the discovery of a 1:2 ratio, as the integer factorisation logic is strict. Future iterations could incorporate a fuzzy GCD or clustering-based approach to handle near-proportionality.

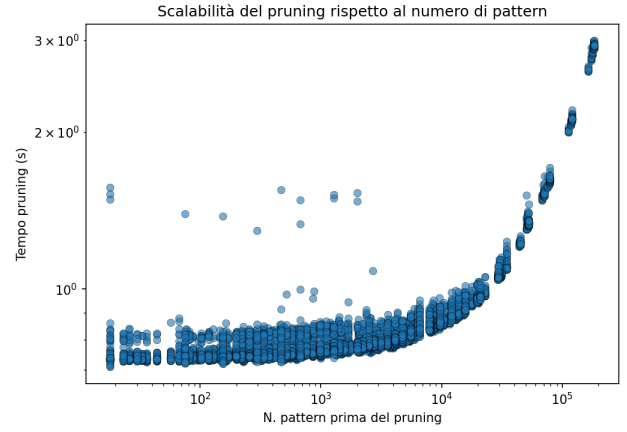


Figure 6: Pruning phase execution time as a function of input pattern count (log-log scale). Time remains sub-second below $\sim 10,000$ patterns and grows toward 2–3 seconds at $\sim 150,000$, consistent with the local quadratic cost of pairwise divisibility checks within each canonical shape group.

- **Strict Proportionality Constraints:** The current model assumes that consumer behaviour scales linearly ($P = \alpha \cdot S$). It does not account for non-linear trends, such as bulk-buy behaviours where product ratios might shift as total volume increases.
- **Absence of Temporal Analysis:** The pipeline processes the dataset as a static snapshot. It does not currently capture seasonal variations or time-based shifts in purchase proportions, which are critical for dynamic retail environments.
- **Local Execution Environment:** All experiments were conducted on a single machine in Spark’s `local[*]` mode. While this validates algorithmic correctness and relative scaling behaviour, it does not replicate the inter-node communication patterns and network overhead of a true distributed cluster. Deployment on a cloud infrastructure such as Azure HDInsight or Amazon EMR would be required to validate the near-linear speedup claim at industrial scale.

6.2 Conclusion

This work has successfully extended the traditional Frequent Itemset Mining (FIM) paradigm by introducing a distributed framework for proportional quantitative analysis. By leveraging the Parallel FP-Growth (PFP) engine, we have demonstrated that it is possible to move beyond simple binary associations to uncover the underlying structural ratios of consumer behaviour without sacrificing computational scalability.

Our implementation introduced three key technical contributions:

- **Bijective Token Indexing:** A transformation layer that allows the PFP distributed engine

to process quantitative tuples while maintaining compatibility with standard prefix-tree structures. [19]

- **Structural Decomposition:** A mathematical approach, anchored in the Prefix Path Property of FP-trees, that utilises integer factorisation to detach a pattern’s irreducible Canonical Shape from its scalar magnitude. [20]
- **Topological Synthesis:** A post-processing phase that applies Transitive Modulo Pruning to eliminate redundant informational pathways, transforming a flat list of patterns into an optimised Directed Acyclic Graph (DAG). [21]

The performance analysis, conducted on the Dunnhumby *The Complete Journey* dataset, confirms that the mining phase achieves near-linear scaling with respect to dataset size, while the Proportional Graph Construction phase adds at most 6% overhead to total execution time. Pipeline correctness was verified end-to-end through a synthetic worked example exercising both the multi-parent convergence and linear scaling topologies. By providing an Atlas of Proportionality rather than a simple list of popular items, this system empowers businesses with deeper insights for automated bundle recommendations and targeted inventory management, effectively bridging the gap between big data mining and actionable business intelligence.

References

- [1]
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